

A close-up photograph of a lizard, possibly a spiny-tailed lizard, resting on a light-colored, textured rock. The lizard's body is covered in brown and tan scales with darker, wavy patterns. Its head is turned slightly to the left, and its eyes are visible. The background is a blurred continuation of the rocky surface.

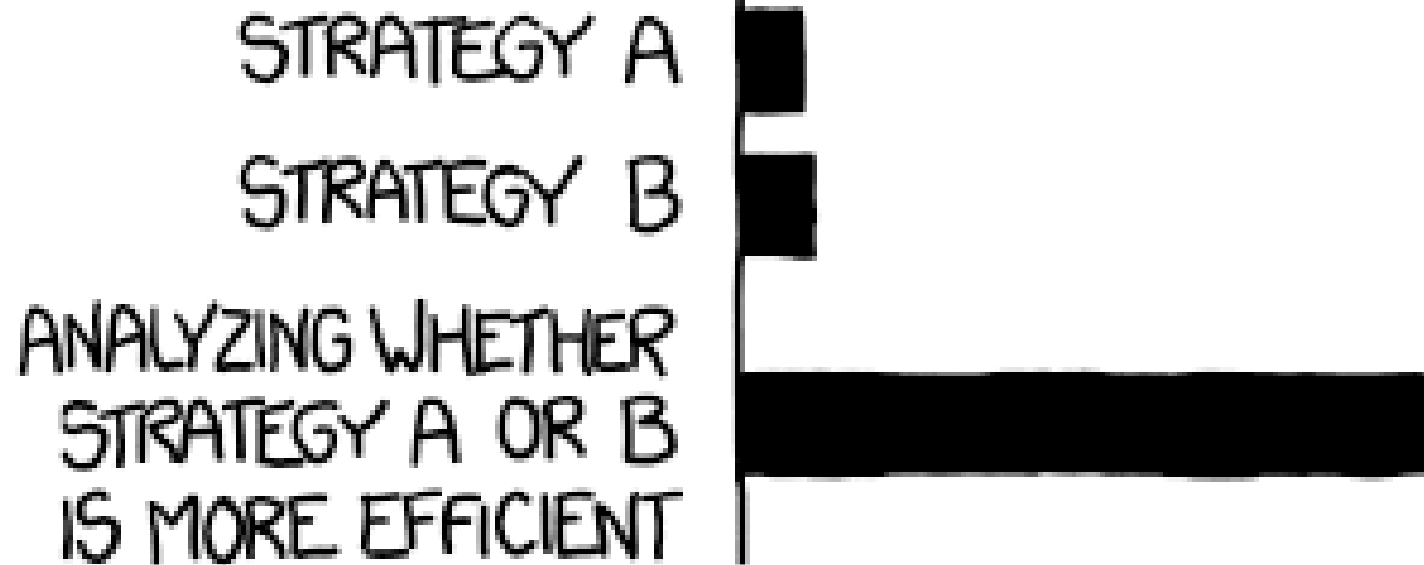
Alternative strategies and trade-offs

IB 431 – Behavioral Ecology
Fall 2025
October 24th 2025

Outline

- I. Alternative strategies
- II. Behavioral trade-offs
- III. Predation / foraging trade-off game
- IV. Discuss Thaker et al 2009

TIME COST



THE REASON I AM SO INEFFICIENT

Alternative strategies

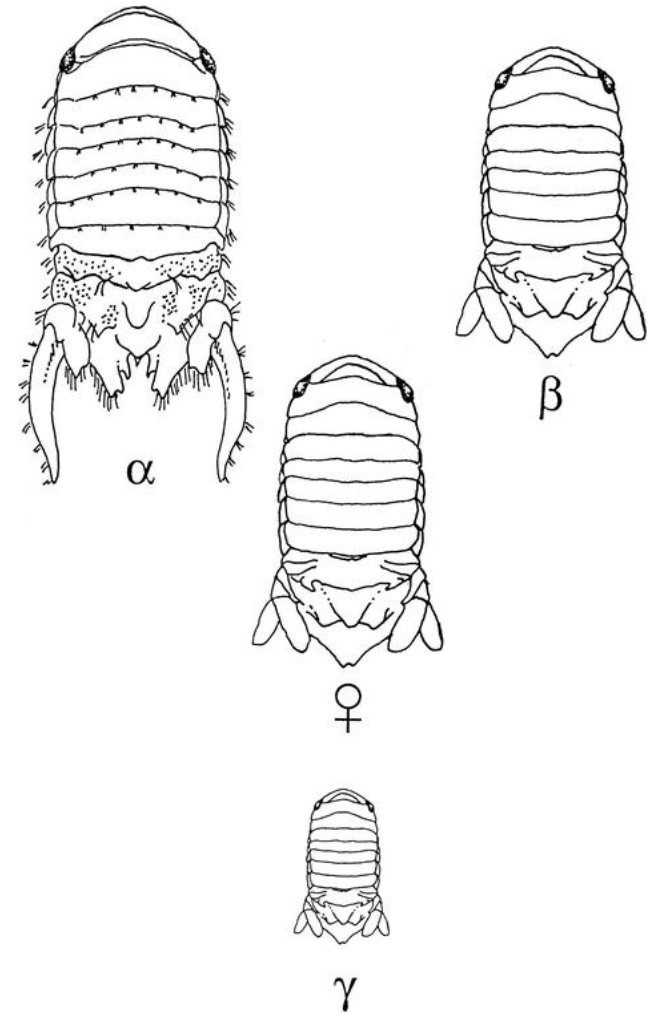
- There may be multiple solutions to a problem
- Frequency-dependent selection can maintain alternative phenotypes
- Differences in female preference can also generate alternative phenotypes



Strawberry poison dart frog (*Oophaga pumilio*)

Alternative mating strategies

- Many examples of dominant and “sneaker” males
 - Marine isopods
 - Bluegill sunfish
 - Atlantic salmon
 - Tree frogs
 - Elephant seals
 - Cuttlefish



Natterjack toads

- Some males will remain silent and use other males' calls to find females
- Interaction with body size: small males tend to use this satellite strategy



Side-blotched lizards

- Orange: aggressive, defend large territories
- Blue: defend one female
- Yellow: males mimic females to sneak matings
 - Orange > blue
 - Blue > yellow
 - Yellow > orange



Alternative foraging strategies

- Producers vs. scroungers



Alternative foraging strategies

- Producers vs. scroungers
- Ambush predation vs. active pursuit
 - Fire salamanders (*Salamandra salamandra*) shift to active foraging when in cave environments



Alternative foraging strategies

- Producers vs. scroungers
- Ambush predation vs. active pursuit
- Feeding behaviors
 - Stabbing vs. hammering to open shells
 - Related to beak morphology



Oystercatcher (*Haematopus ostralegus*)

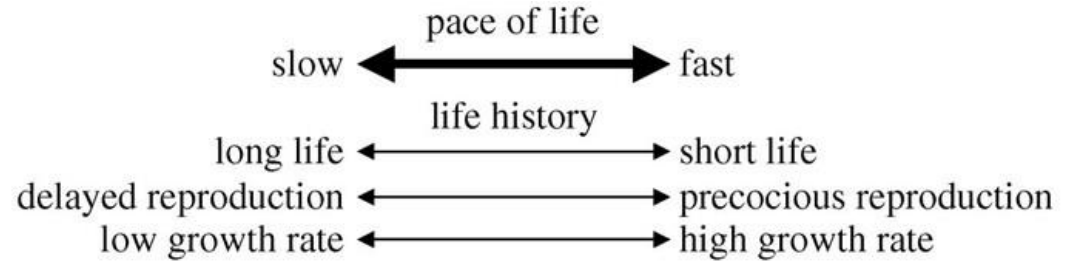
Alternative dispersal strategies

- Blackbirds migrate in winter
- Some birds remain as “residents” and do not migrate
- Residents must deal with winter conditions but are able to maintain their territory and invest less energy preparing for migration



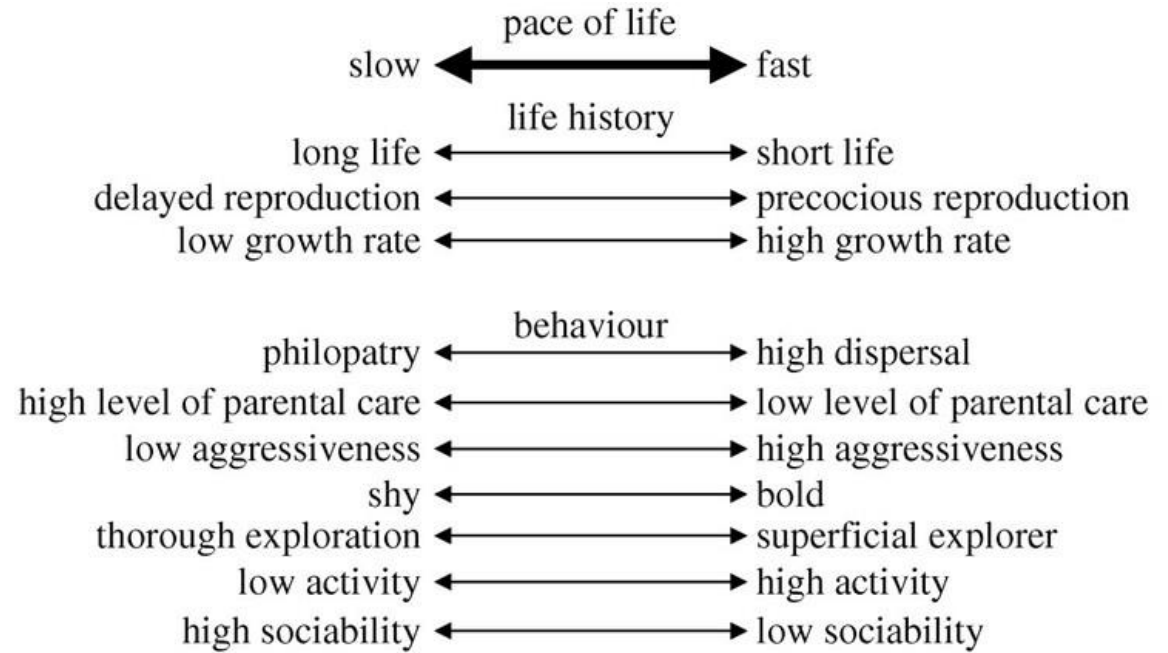
Pace of life syndrome

- Suite of correlated traits
- Generally – fast vs. slow strategies (r vs. K selection)



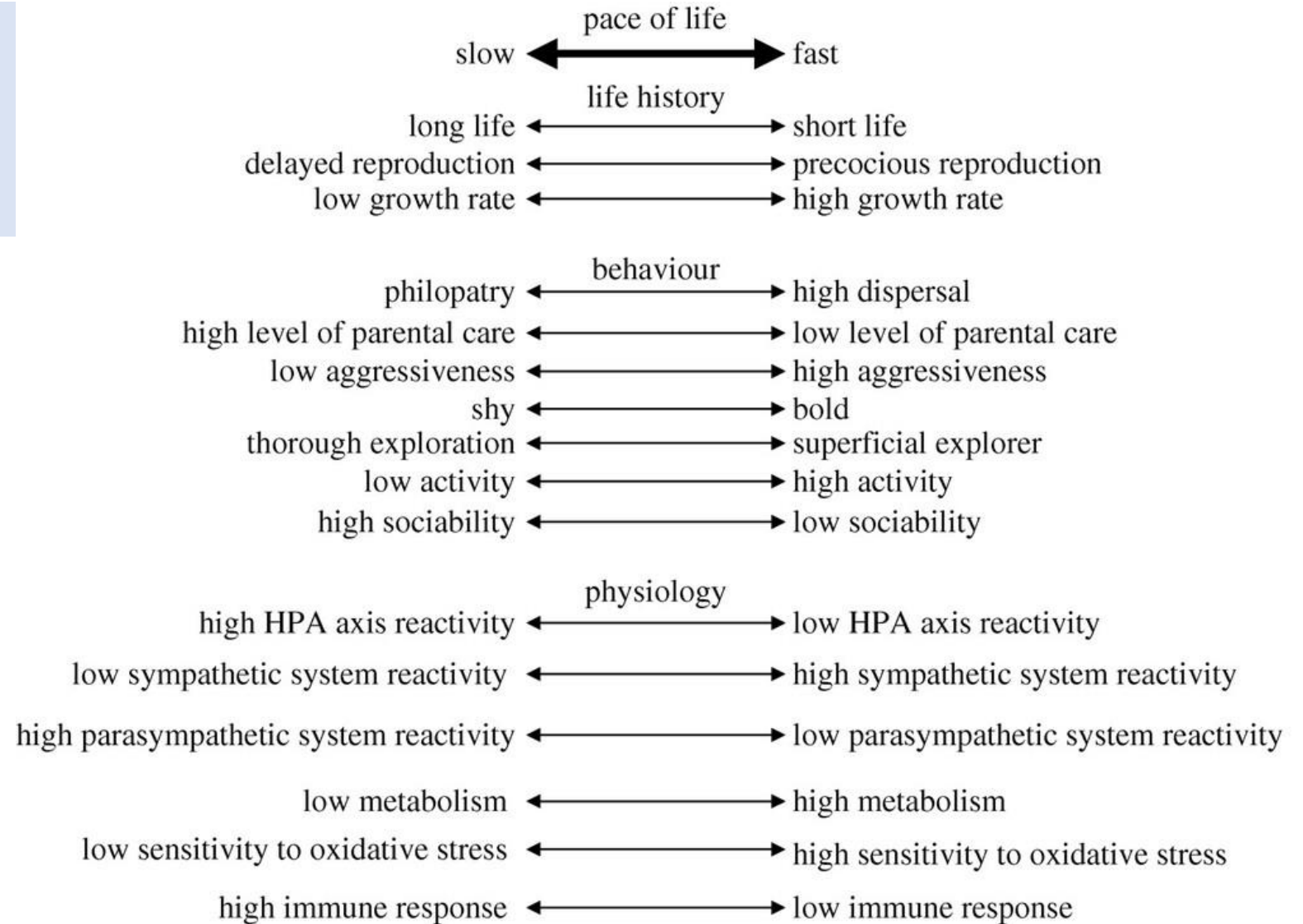
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Pace of life syndrome

- Suite of correlated traits
- Generally – fast vs. slow strategies (r vs. K selection)



Pace of life syndrome

- Evidence for this theory remains mixed but there are compelling examples
- Often mediated by environmental stress

Pace of life syndrome

- Evidence for this theory remains mixed but there are compelling examples
 - Guppies facing high predation have fast POL – develop faster, reproduce sooner, and have increased risk-taking tendencies



Reznick et al 1998

Pace of life syndrome

- Evidence for this theory remains mixed but there are compelling examples
 - When food availability is low, red squirrels become more aggressive, increase activity level, and reproduce at an earlier age



Boon et al 2008

How do alternative strategies develop?

- Reversible vs. irreversible

How do alternative strategies develop?

- Reversible vs. irreversible
 - Reversible: Cooperative vs. cheating strategies
 - Cleaner wrasse will shift strategy depending on recent social context, resource availability, type of client fish present



How do alternative strategies develop?

- Reversible vs. irreversible
 - Irreversible: Dung beetle horned vs hornless males
 - Driven by combination of maternal effects, nutrition, and body size

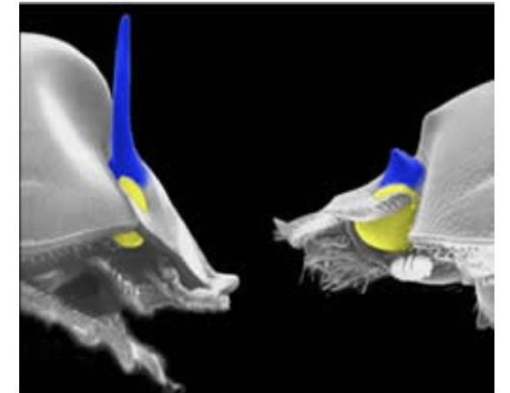
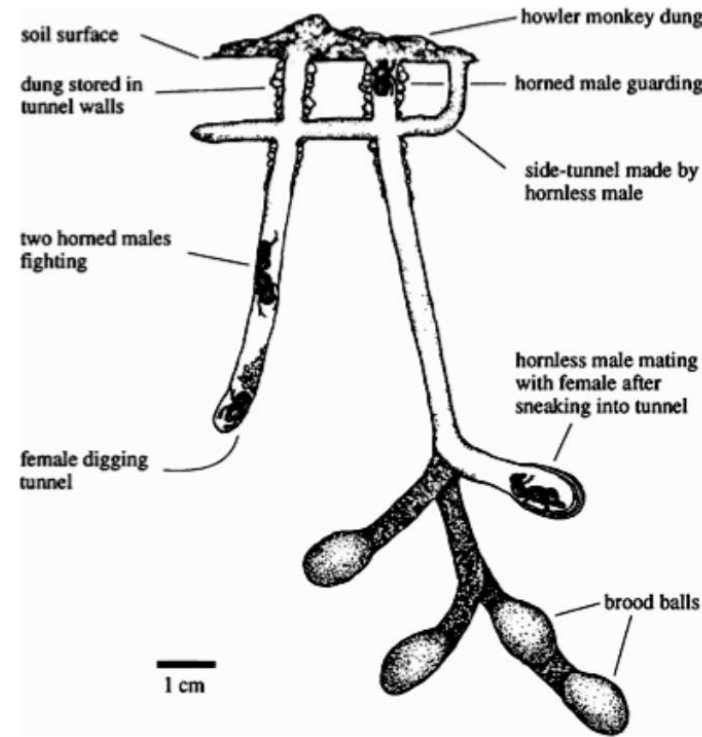


Fig. 1 Biology of *Onthophagus acuminatus* illustrating the two reproductive tactics used by males. Horned males fight to guard possession of tunnels containing females ("guarding"), and males able to successfully guard tunnels mate repeatedly with occupant females. Hornless males sneak into tunnels by digging horizontal side-tunnels that intercept primary tunnels underground ("sneaking"). In this way hornless males sometimes enter main tunnels beneath the guarding male and mate with the female undetected

Hormones may shape development of alternative strategies

- Activational effects: short-term, reversible effects that occur in the fully developed organism
 - Pre-maternal mice will attack pups
 - Treatment with oxytocin induces maternal behavior



Hormones may shape development of alternative strategies

- Activational effects: short-term, reversible effects that occur in the fully developed organism
 - Dolly the cloned sheep cared for surrogate lamb thanks to oxytocin treatment



Hormones may shape development of alternative strategies

- Organizational effects: long-term, irreversible effects on development (e.g., brain, ovary) that occur early in life
 - Critical periods for development early in life
 - Neonatal testosterone exposure led to adult male-typical behavior in female rats



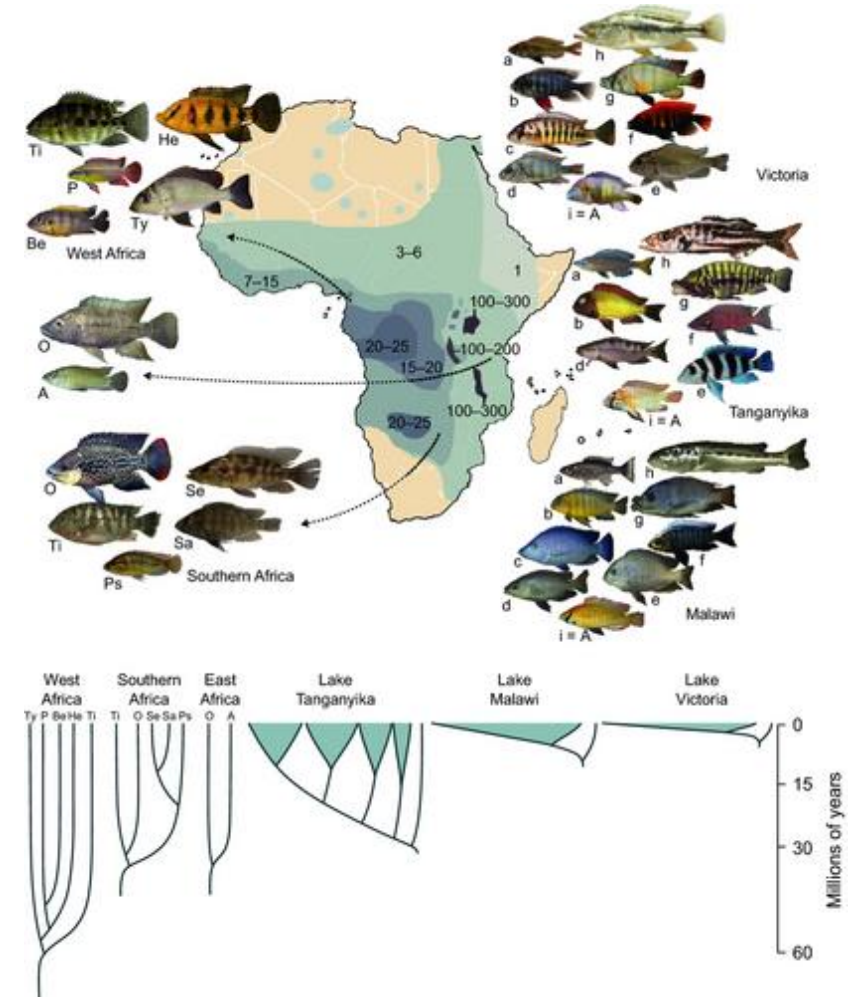
Alternative strategies and development

- Reversible alternative strategies tend to be driven by activation effects
- Irreversible alternative strategies tend to be driven by organization effects



Alternative strategies can lead to speciation

- Differences in mating and foraging strategy likely played role in cichlid radiation in Africa



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Behavioral trade-offs

- Behavioral decisions where animals must choose to increase one aspect of fitness at the cost of another
- Can you think of any potential behavioral trade-offs? Discuss with your table for ~3 minutes.



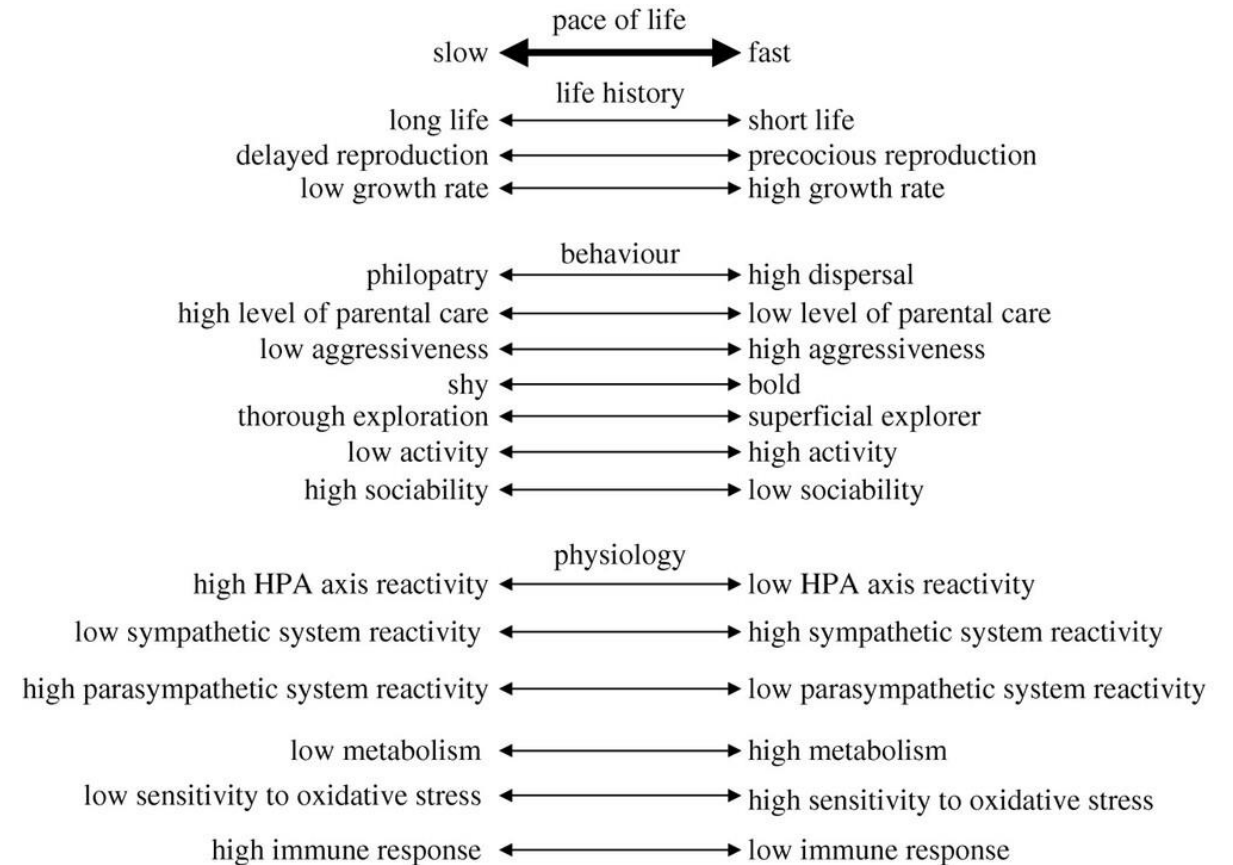
Examples of behavioral trade-offs

- Predator vigilance vs. doing something else
 - Foraging
 - Courting mates
 - Defending territory
 - Resting
 - Thermoregulation



Examples of behavioral trade-offs

- Current vs. future reproduction
 - Fast vs. slow POL syndromes
- Growth/resource acquisition vs. reproductive effort
- Survival vs. reproductive effort



Examples of behavioral trade-offs

- Exploration-exploitation trade-off
 - Invest energy in exploiting a known resource or exploring for a better resource



Examples of behavioral trade-offs

- Social groups
 - Trade-off between benefits of cheating in cooperative group vs. costs of not cooperating
 - Benefits of social living (communication, safety, etc.) vs costs (increased disease transmission, aggression)



Examples of behavioral trade-offs

- Speed vs. accuracy in decision making
 - Ants that take more time relocating nests are more accurate
 - Archerfish trained to shoot at targets to get food; faster individuals were less accurate





More videos 

Preview

Love
Nature



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Meerkat Manor 2

Congratulations!

You are now meerkats

You are a small, highly social group of mongoose living in South Africa

Drongos torment you

Times are hard, and you must do your best to survive



What now?

You must obtain enough food (points) to survive the coming year

Each round, you may either forage or serve as sentinel

Foraging nets you 1 point

Serving sentinel nets you 0 points



Beware!

A hawk stalks the desert
(and loves to eat meerkats)!

The hawk will *always* eat 1
player

Hawk will eat a random
player from the table with
the fewest sentinels (not
safe even if you're a
sentinel)

If eaten, you're out!



End Game.

We will play a number of rounds

Player with the most food (points) will be crowned the greatest meerkat

The survival of your table is secondary to your own!

Good luck!



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